SELF-BIAS

JFET DC BIASING

prepared by:

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Electronics Engineer

GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

50%

WEEKLY

50%

EXAM

100%

LABORTORY

PENALTY

0

ABSENT

3

45-min LATE

5 max

15-min LATE

TOPIC OUTLINE

Self-Bias

- Gate-to-Source Loop
- Drain-to-Source Loop
- Transconductance Curve

SELF-BIAS

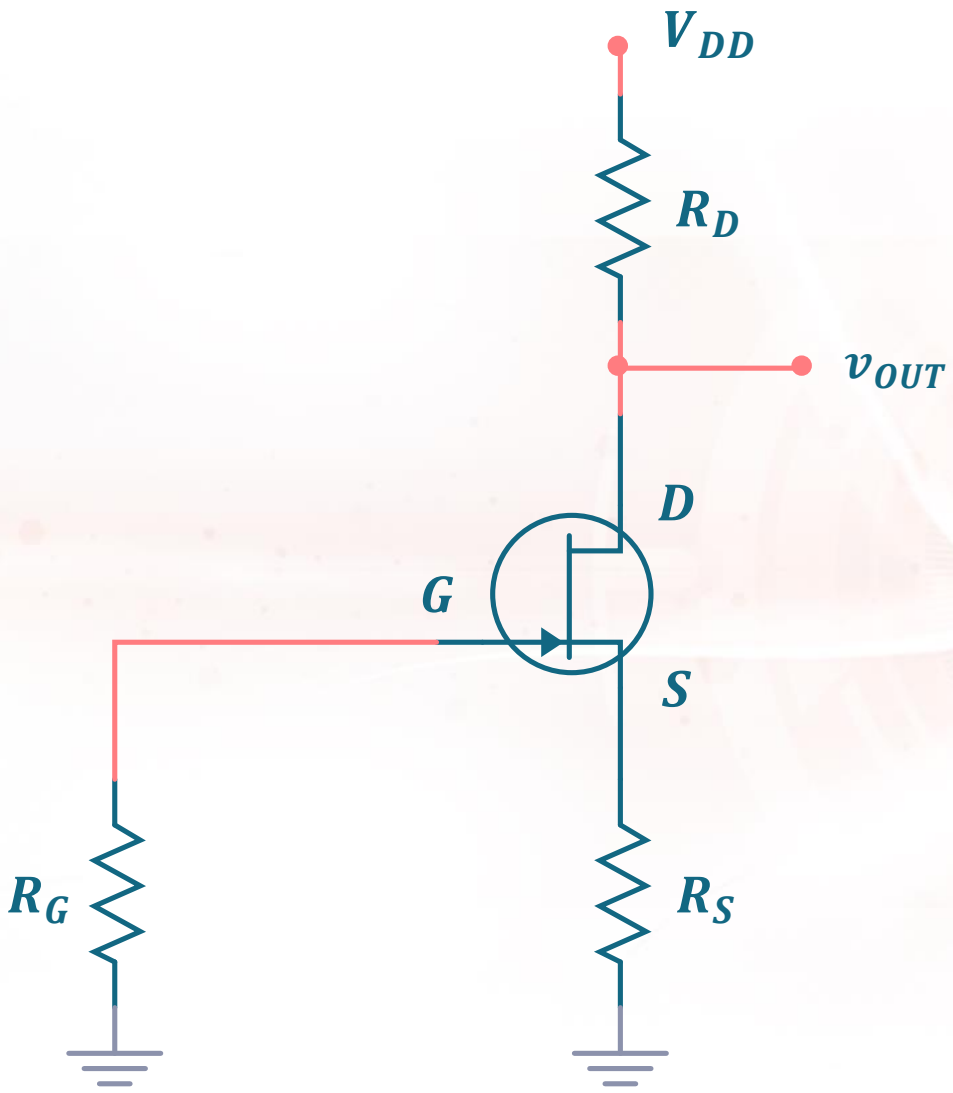
GENERAL RELATIONSHIPS

$$i_D = i_{DSS}\left(1 - \frac{v_{GS}}{v_P}\right)^2$$

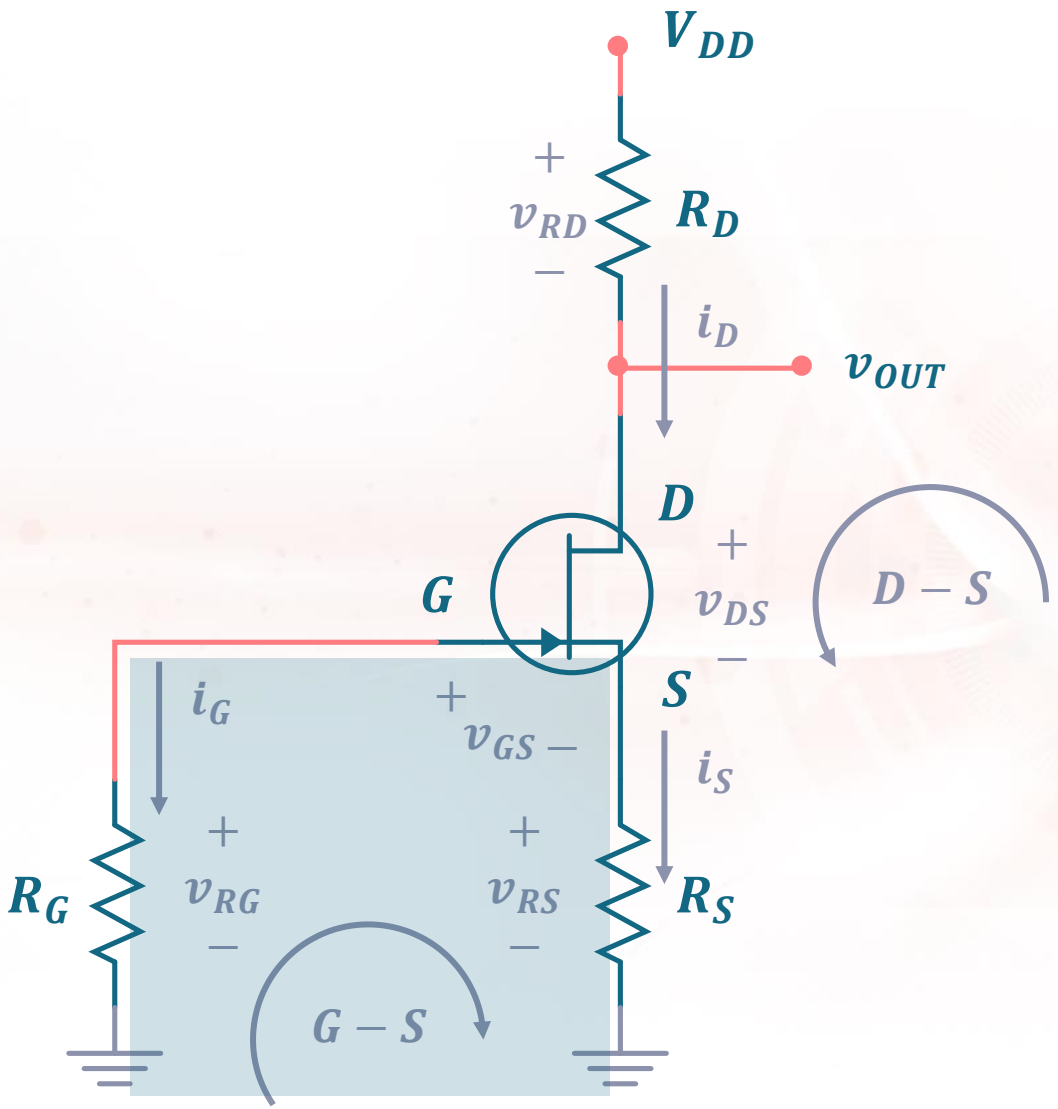
$$i_G \cong 0$$

$$i_D = i_S$$

SELF-BIAS CONFIGURATION



GATE-TO-SOURCE



KVL @ G-S

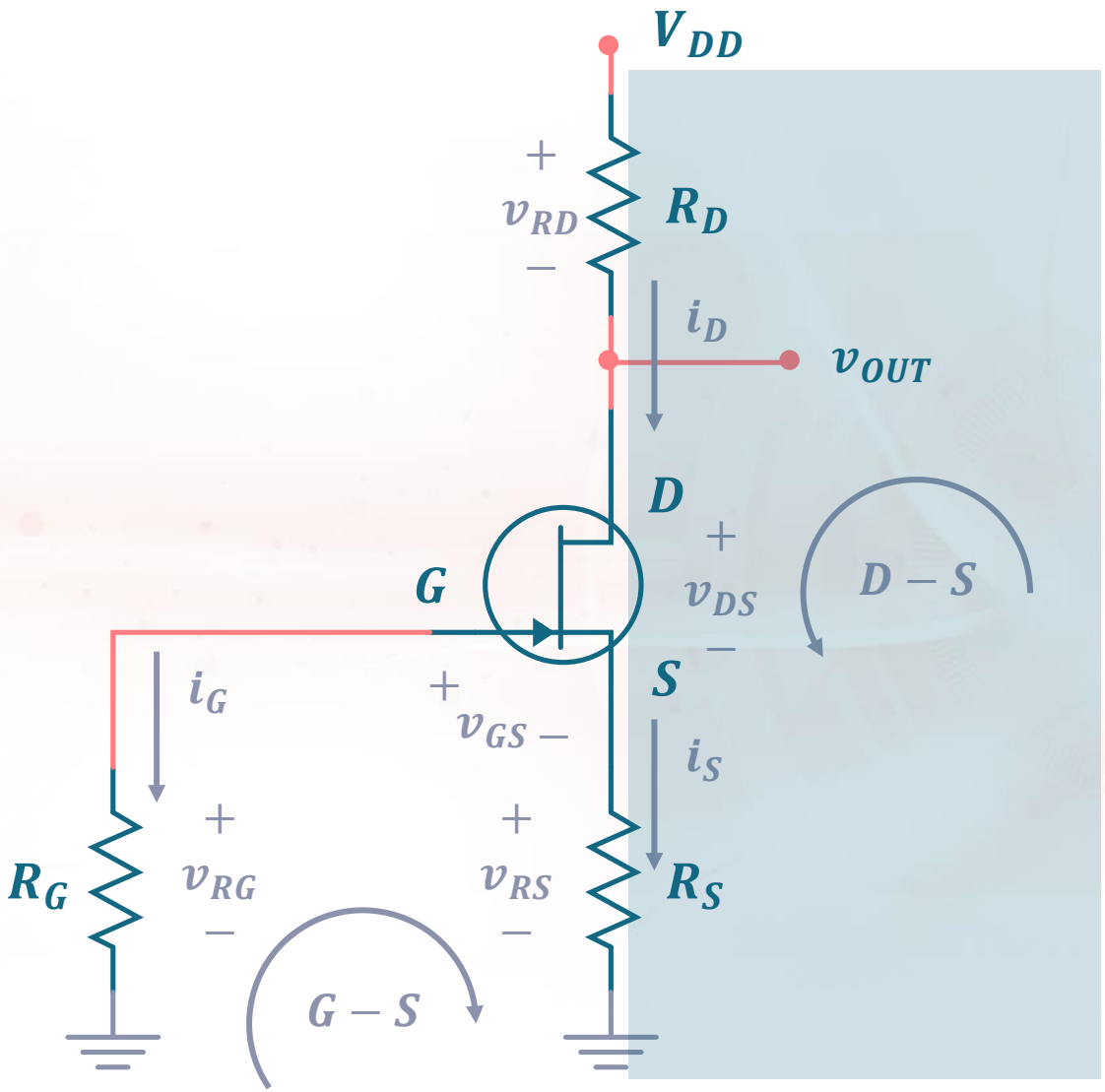
$$-v_{RG} + v_{GS} + v_{RS} = 0$$

$$v_{GS} = v_{RG} - v_{RS}$$

$$v_{GS} = \cancelto{0}{i_G R_G} - i_S R_S \qquad ; \; i_S = i_D$$

$$\boxed{v_{GS} = -i_D R_S}$$

DRAIN-TO-SOURCE



KVL @ D-S

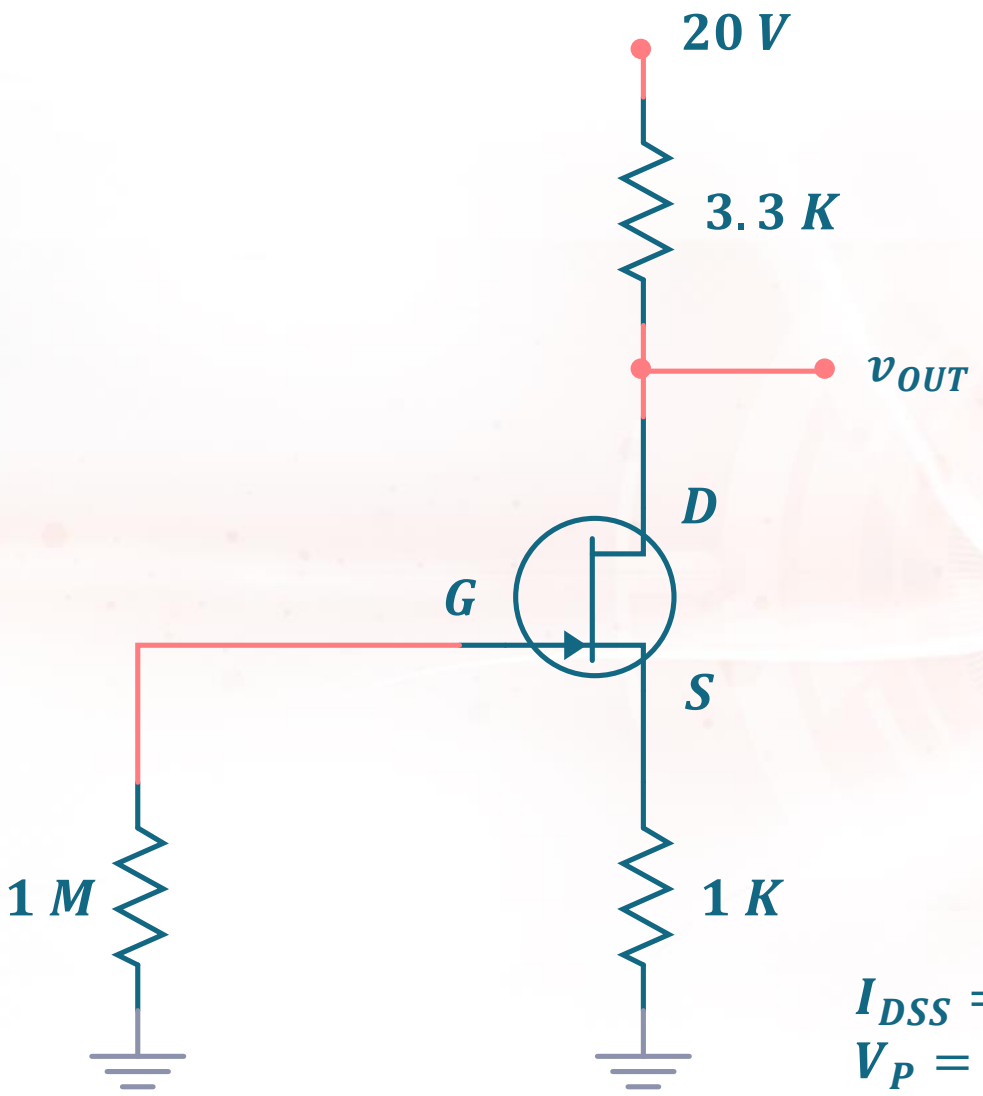
$$-v_{DD} + v_{RD} + v_{DS} + v_{RS} = 0$$

$$v_{DS} = v_{DD} - v_{RD} - v_{RS}$$

$$v_{DS} = v_{DD} - i_D R_D - i_S R_S$$

$$\boxed{v_{DS} = v_{DD} - i_D(R_D + R_S)}$$

EXERCISE



$I_{DSS} = 8\ mA$
$V_P = -6\ V$

For the given network, determine the following :

v_{GSQ}

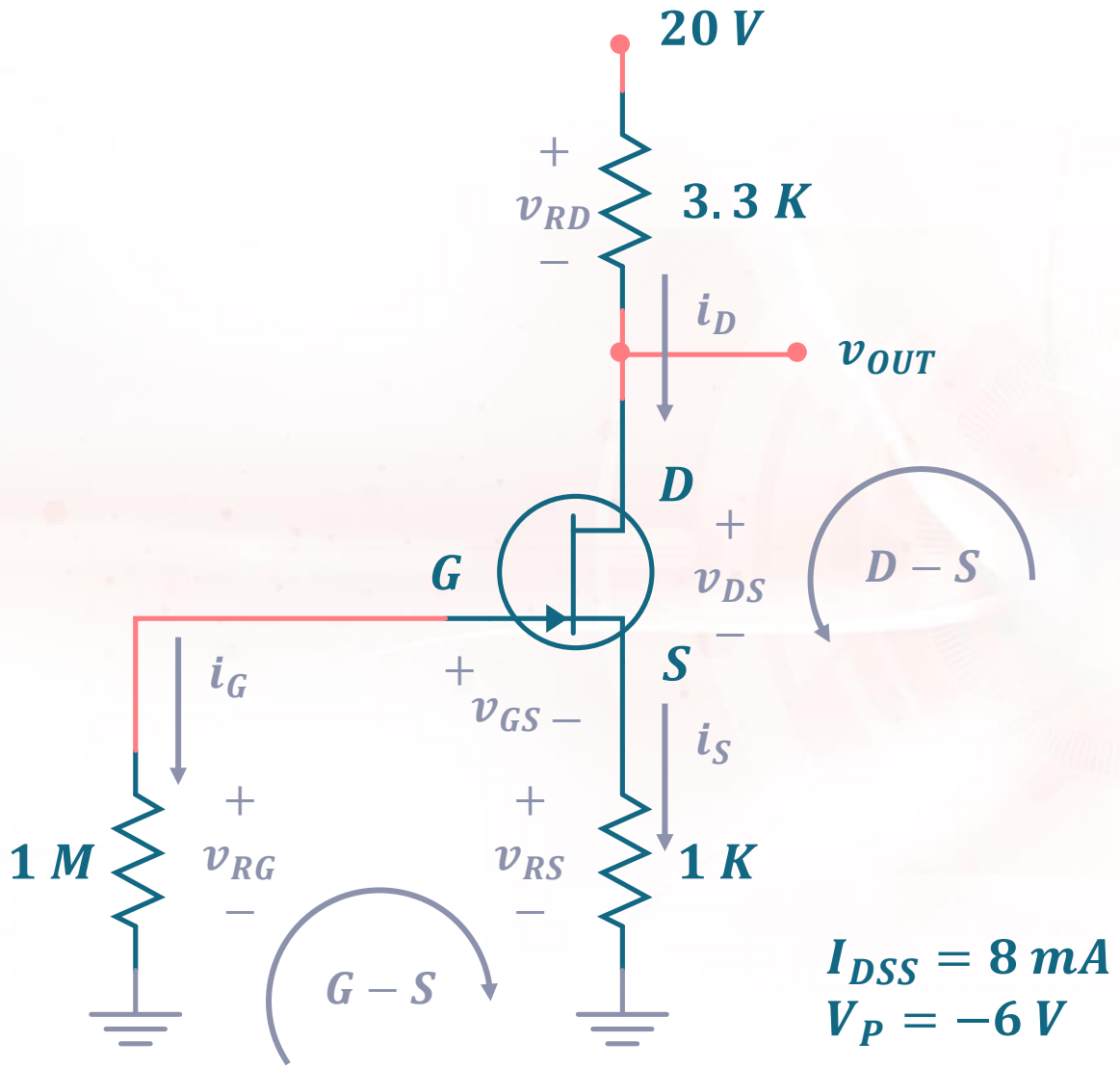
i_{DQ}

v_{DS}

v_S

and **sketch** the transconductance curve.

EXERCISE



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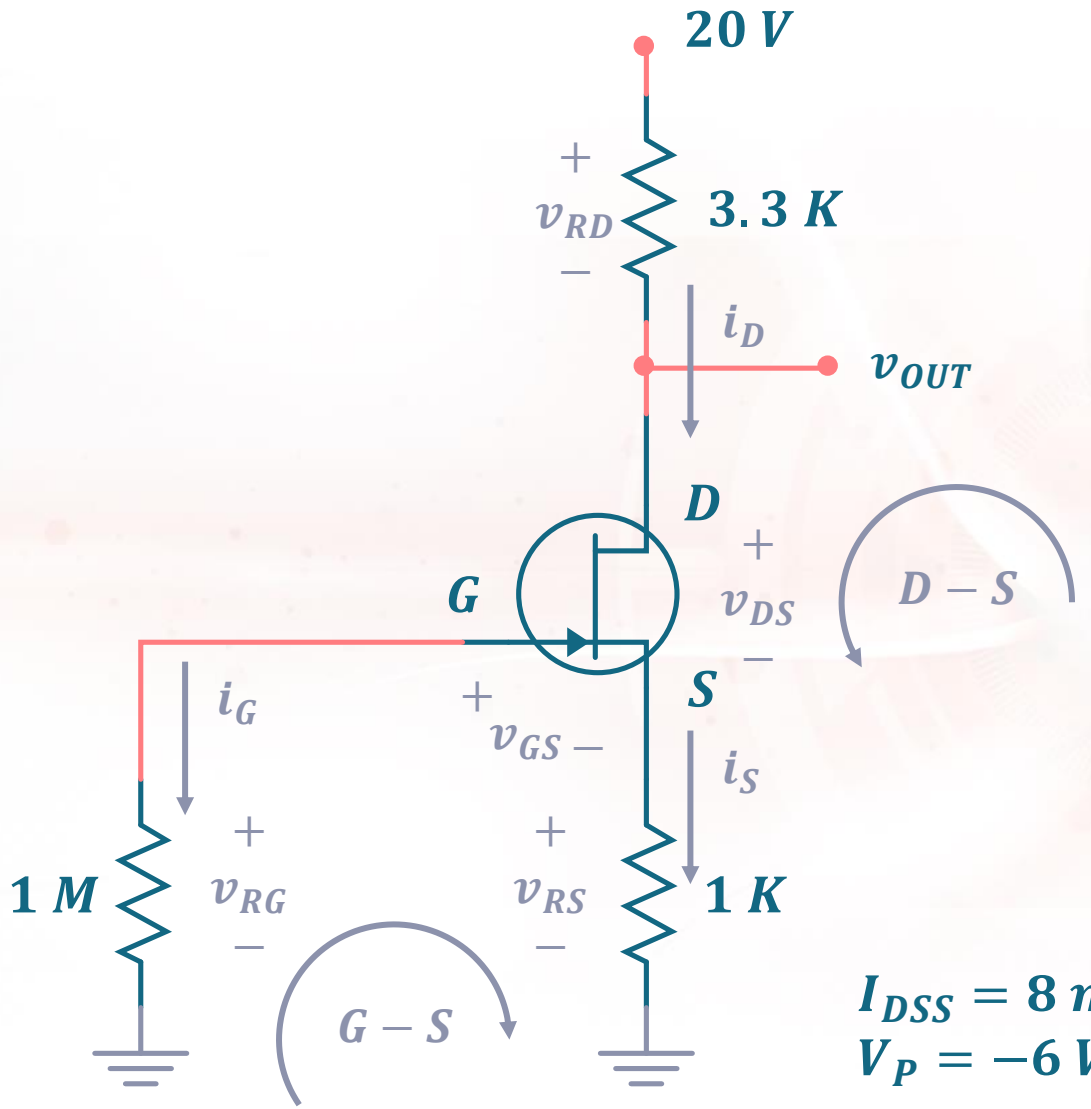


$I_{DSS} = 8\ mA$
$V_P = -6\ V$

KVL @ G-S

$$-v_{RG} + v_{GS} + v_{RS} = 0$$

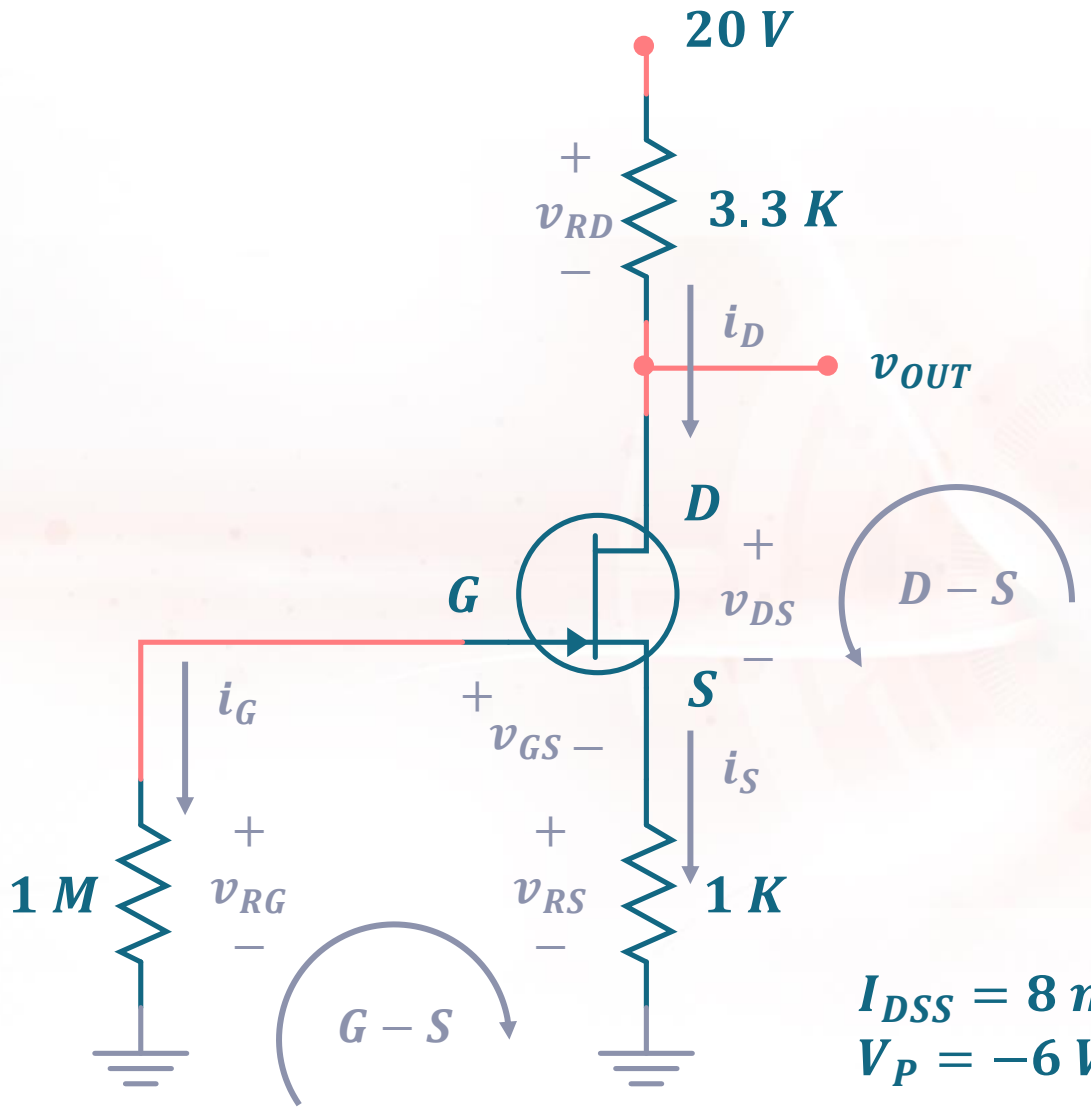
$$v_{GS} = v_{RG} - v_{RS}$$

$$v_{GS} = i_G R_G - i_S R_S$$

$$v_{GS} = -i_D R_S$$

$$v_{GS} = -i_{DSS}\left(1 - \frac{v_{GS}}{v_P}\right)^2 R_S$$

$$v_{GS} = -8\ mA\left(1 - \frac{v_{GS}}{-6}\right)^2 1K$$



$I_{DSS} = 8\ mA$
$V_P = -6\ V$

KVL @ G-S

$$v_{GS} = -8\ mA\left(1 + \frac{v_{GS}}{6}\right)^2 1K$$

$$v_{GS} = -8\left(1 + \frac{v_{GS}}{3} + \frac{v_{GS}^2}{36}\right)$$

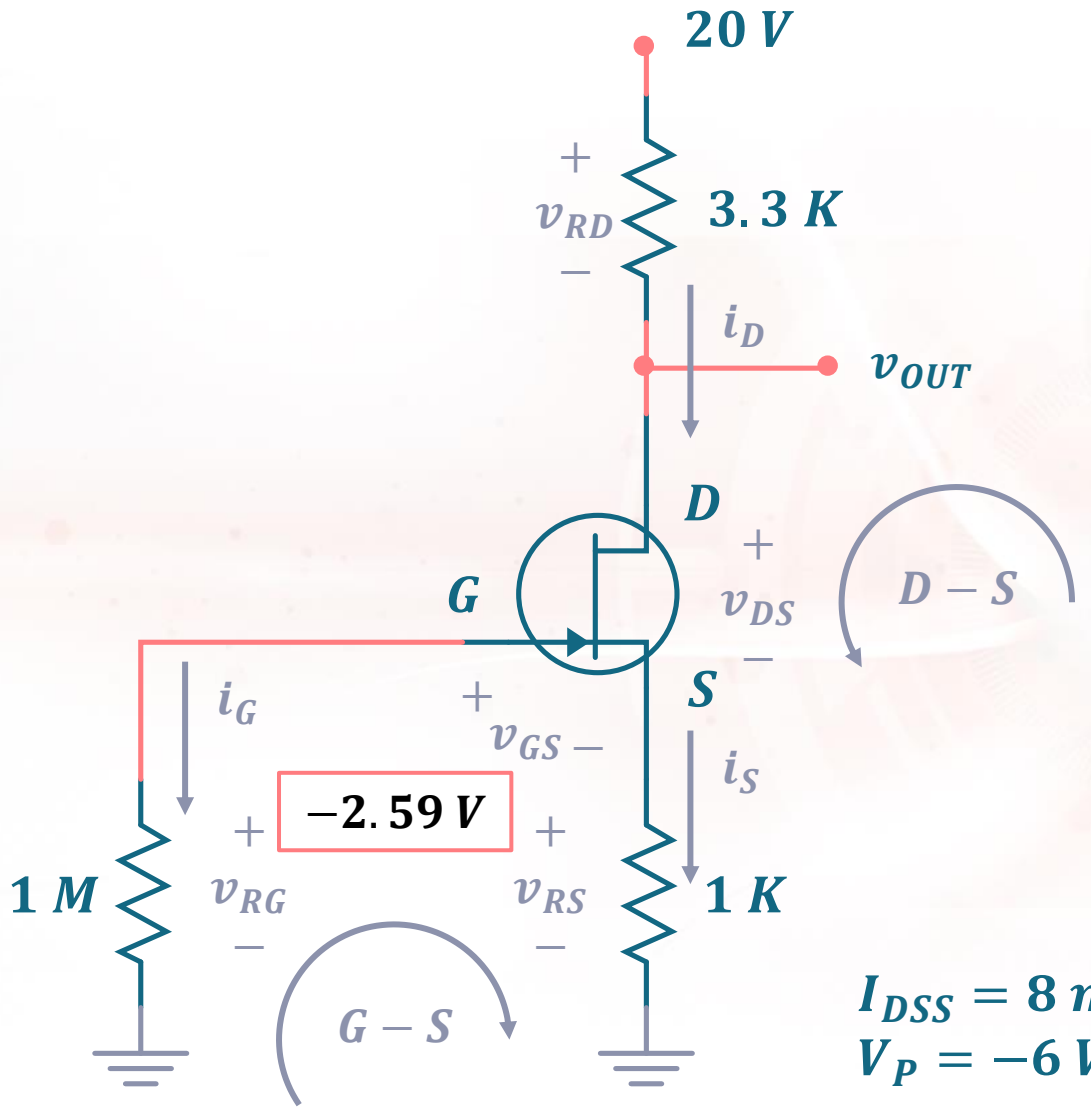
$$v_{GS} = -8 - \frac{8}{3}v_{GS} - \frac{2}{9}v_{GS}^2$$

$$0 = -8 - \frac{11}{3}v_{GS} - \frac{2}{9}v_{GS}^2$$

$$v_{GS} = -13.91$$

$$\boxed{v_{GSQ} = -2.59\ V}$$

ans



$I_{DSS} = 8\ mA$
$V_P = -6\ V$

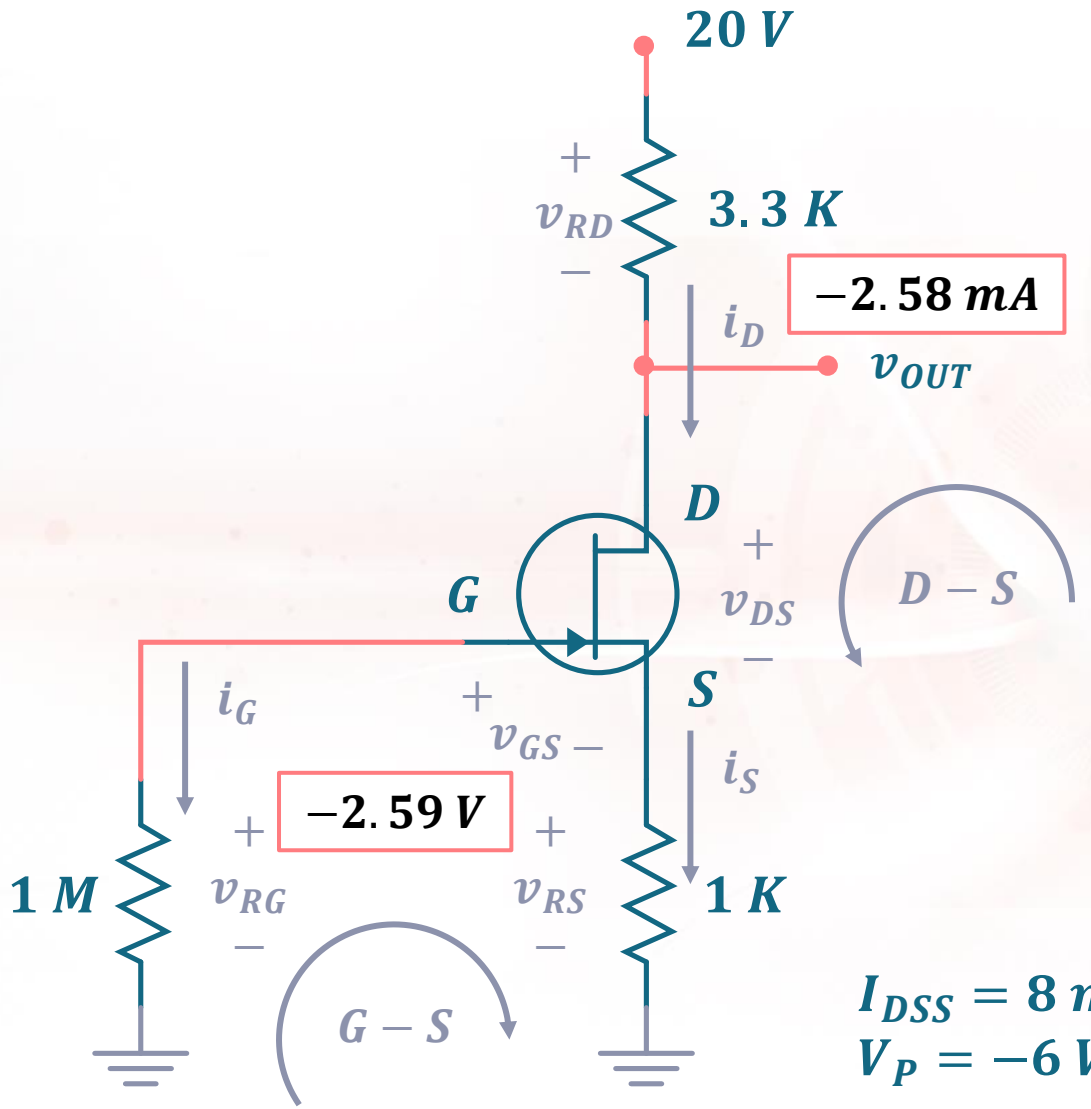
Shockley's Equation

$$i_D = i_{DSS}\left(1 - \frac{v_{GS}}{v_P}\right)^2$$

$$i_D = 8\ mA\left(1 - \frac{-2.59}{-6}\right)^2$$

$$\boxed{i_{DQ} = 2.58\ mA}$$

ans



$I_{DSS} = 8\ mA$
$V_P = -6\ V$

KVL @ D-S

$$-v_{DD} + v_{RD} + v_{DS} + v_{RS} = 0$$

$$v_{DS} = v_{DD} - v_{RD} - v_{RS}$$

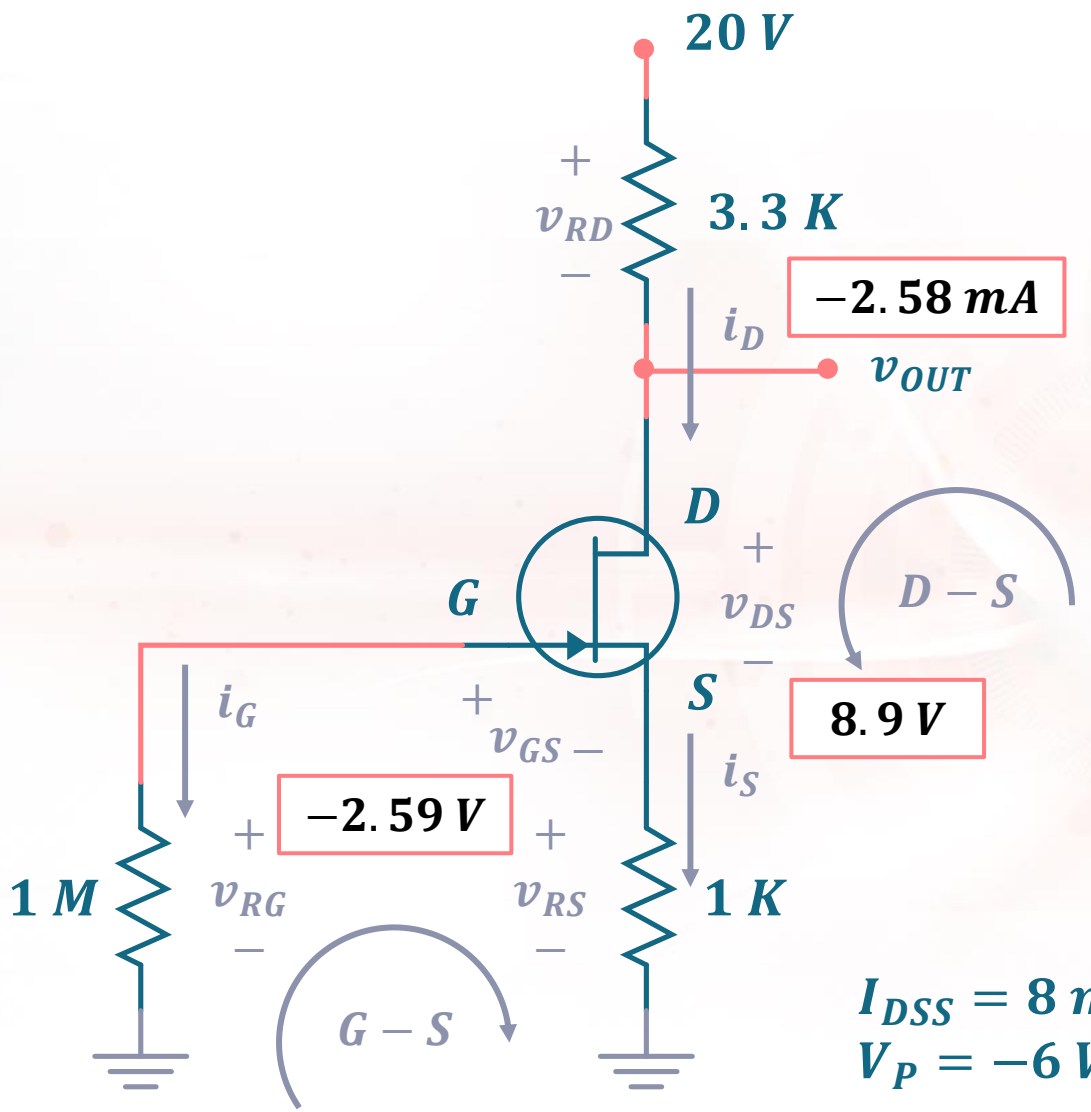
$$v_{DS} = v_{DD} - i_D R_D - i_S R_S$$

$$v_{DS} = v_{DD} - i_D(R_D + R_S)$$

$$v_{DS} = 20 - 2.58\ mA(3.3\ K + 1\ K)$$

$$\boxed{v_{DSQ} = 8.9\ V}$$

ans



$I_{DSS} = 8\ mA$
$V_P = -6\ V$

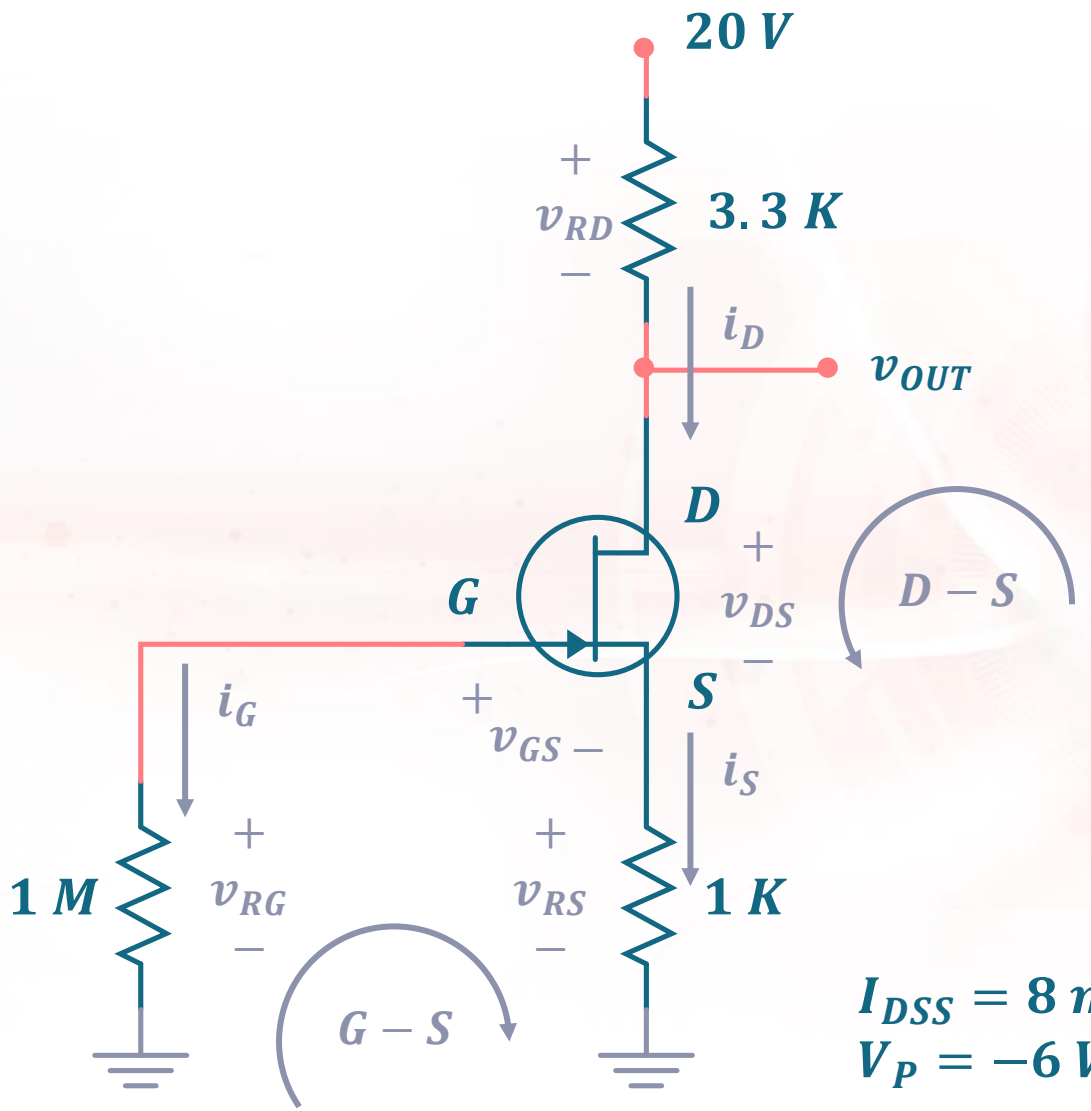
Source Voltage

$$v_S = i_D R_S$$

$$v_S = 2.58\ mA(1K)$$

$$\boxed{v_S = 2.58\ V}$$

ans



$$I_{DSS} = 8\ mA$$
$$V_P = -6\ V$$

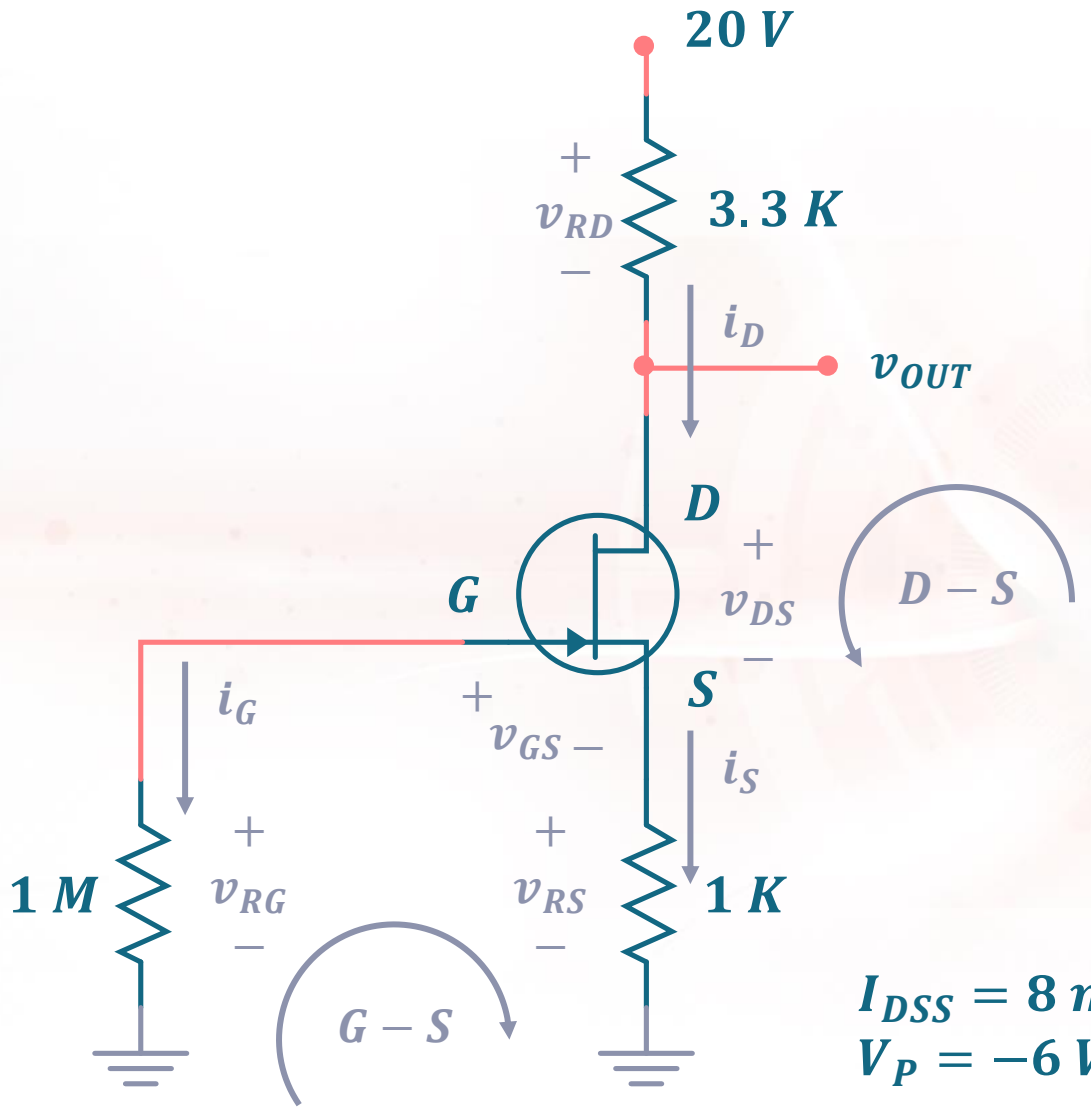
Transconductance Curve

$$i_D = \frac{1}{4} i_{DSS}\Big|_{v_{GS}=0.5v_P}$$

$$i_D = \frac{1}{4}(8\ mA)\Big|_{v_{GS}=-3\ V}$$

$$\boxed{i_D = 2\ mA\Big|_{v_{GS}=-3\ V}}$$

ans



$I_{DSS} = 8\ mA$
$V_P = -6\ V$

Transconductance Curve

$$v_{GS} = 0.3v_P\Big|_{i_D=0.5i_{DSS}}$$

$$v_{GS} = 0.3(-6)\Big|_{i_D=4\ mA}$$

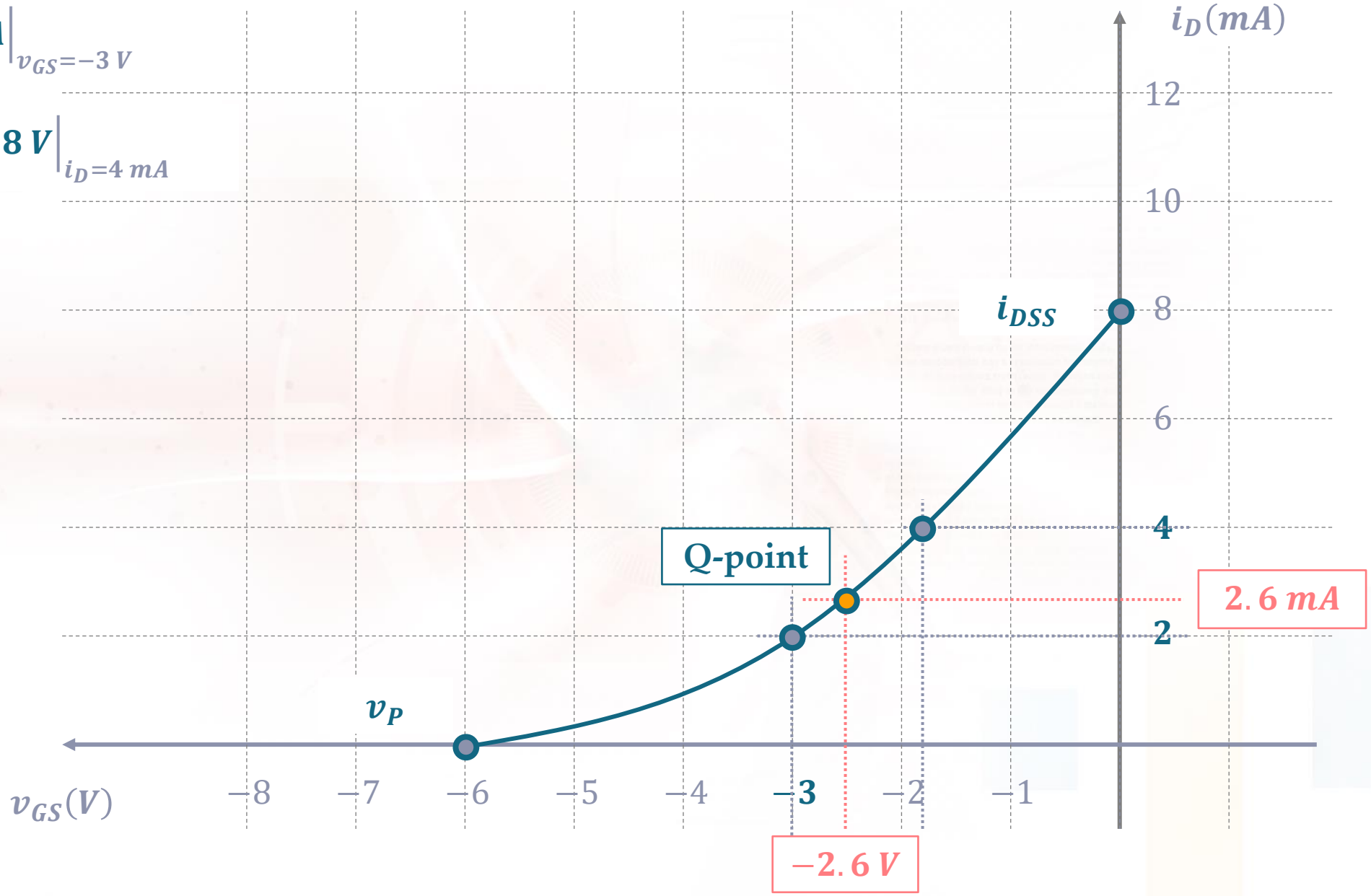
$$\boxed{v_{GS} = -1.8\ V\Big|_{i_D=4\ mA}}$$

ans

TRANSCONDUCTANCE CURVE

$$i_D = 2\ mA\Big|_{v_{GS}=-3\ V}$$

$$v_{GS} = -1.8\ V\Big|_{i_D=4\ mA}$$



LABORATORY